



Cosmic rays at the epoch of reionization

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Resumen / Uno de los problemas abiertos más importantes de la Cosmología actual es el de las fuentes responsables de la ionización y calentamiento del medio intergaláctico (MIG) durante el Amanecer Cósmico. La radiación de las primeras galaxias no sería suficiente para mantener la ionización del medio a gran escala, por lo que se han propuesto agentes de ionización complementarios, entre ellos los rayos cósmicos producidos en dichas galaxias. En este artículo mostramos nuestros primeros resultados sobre la ionización del MIG producida por cascadas electromagnéticas iniciadas por electrones, en un escenario estándar de reionización. Utilizando métodos numéricos simulamos el transporte de las partículas que componen las cascadas. Luego, calculamos la tasa de ionizaciones y la distribución de energía entre las partículas y el MIG al propagarse dichas cascadas. Nuestros resultados refuerzan y extienden aquellos previamente obtenidos por otros autores, en el sentido de que la contribución de la componente electrónica de rayos cósmicos a la ionización del MIG ocurre en dos etapas. Electrones con energía $\lesssim 10$ keV se enfrían completamente dentro de una distancia del orden de los miles de kiloparsecs, ionizando el medio. Por otro lado, electrones de energías $\gtrsim 100$ keV transportan la misma a distancias mas allá de un megaparsec. Además, mostramos por primera vez que los fotones secundarios producto de cascadas iniciadas por electrones de alta energía podrían contribuir significativamente a la ionización del MIG. Este resultado implica una tasa de ionización considerable en el MIG lejano que aún no ha sido explorada en detalle.

Abstract / One of the most important open problems in modern cosmology is that of the sources responsible for the ionization and heating of the intergalactic medium (IGM) during the Cosmic Dawn. Radiation from early galaxies would not have been enough to maintain the ionization of the IGM on a large scale. For this reason different complementary ionizing agents have been proposed, such as the cosmic rays produced by those galaxies. In this work we present our first results on the ionization of the IGM produced by electron-initiated electromagnetic cascades (EMC), for a standard reionization scenario. Using numerical methods we simulated the transport of the particles that compose the cascades. Then, we calculated the ionization rate and the energy distribution among the particles and the IGM as the cascades propagate through the latter. Our results reinforce and extend those of other authors, in the sense that the contribution of the electronic component of the cosmic rays to the ionization of the IGM occurs in two stages. Electrons with energies below 10 keV cool down completely within a distance to the source on the order of thousands of kiloparsecs, ionizing the medium in the process. On the other hand, electrons with energies above 100 keV transport most of their energy to distances beyond the megaparsec. Also, we show for the first time that secondary photons originated in cascades initiated by high-energy electrons could contribute significantly to the ionization of the IGM. This result implies a sizeable ionization rate in the deep IGM that is yet to be explored in detail.

Keywords / astroparticle physics — cosmic rays — dark ages, reionization, first stars

1. Introduction

With the birth of the first sources of stellar radiation at $z \sim 20$ (Peacock, 1999), the thermodynamic state of the primordial IGM started to change. As photons interacted with the neutral hydrogen and helium atoms, the IGM began to suffer a global phase transition characterized by a progressive increase of its ionization fraction. The period during which this transition occurred is known as the Epoch of Reionization (EoR), and the latest observations indicate that it would have ended by $z \sim 5.7$ (Fan et al., 2006; Becker et al., 2015).

It is currently understood that UV radiation emitted by the first stars was one of the main agents responsible for the reionization of the Universe (e.g. Robertson

et al., 2010). However, evidence suggests that the escape fraction of UV photons would not have been high enough for them to fully ionize the deep IGM, far from the sources (Heckman et al., 2001; Ferrara & Loeb, 2013; Mitra et al., 2013; Izotov et al., 2016).

For these reasons, complementary sources and ionizing agents have been proposed. Some authors studied active galactic nuclei as candidates for heating and ionizing the IGM, although the demography of this objects at high redshift is still uncertain (e.g. Fan et al., 2001; Cowie et al., 2009; Madau & Haardt, 2015). Other authors have considered the contribution of X-ray photons originated from X-ray binaries (Mirabel et al., 2011; Fragos et al., 2013; Jeon et al., 2014; Xu et al., 2014; Sazonov & Khabibullin, 2017), but some works suggest

that their effects on the state of the IGM would have been marginal at best (e.g. Madau & Fragos, 2017).

Cosmic rays (CR) that escaped from galaxies have also been considered as alternative ionizing agents. Subatomic particles could have been accelerated in early supernova explosions, or inside the highly collimated jets associated to Pop III microquasars, transporting energy far away from their sources (Heinz & Sunyaev, 2002; Sotomayor Checa & Romero, 2019). As they travel, CRs can interact with the matter, radiation and magnetic fields in the IGM, and deposit a significant fraction of their kinetic energy into it, favoring the reionization process.

Previous works have explored the contribution of CR electrons leaking from galaxies (Tueros et al., 2014; Leite et al., 2017; Douna et al., 2018). In this article, we improve upon these works taking into account cooling mechanisms not considered by these authors and expanding the electron energy range. We focus on the energy exchange between the electronic component of CRs and the IGM, and identify the most relevant channels of interaction through which electrons could affect the thermodynamic state of the medium during the EoR. To this aim, we designed and ran a set of Monte Carlo simulations of the injection of energy in the IGM through EMC, in order to estimate their contribution to the reionization process.

2. Method

The efficiency of the cooling processes that a CR electron could suffer while traversing the IGM strongly depends on the density of its ambient fields. To compute the energy losses produced by the interactions of CRs with the early IGM, we considered the latter as a partially ionized hydrogen gas. We analyzed the characteristic timescales of the interactions for a standard reionization scenario characterized by two parameters: the redshift z and the neutral hydrogen ionization fraction f_{ion} . We adopted the typical values at the beginning of the EoR ($z = 10$ and $f_{\text{ion}} = 10^{-4}$) since most models favor a late starting and rapid process (Greig & Mesinger, 2017).

To model the partially ionized hydrogen gas we considered the following four ambient fields: Two matter fields with densities $\rho_e = 0.025 \text{ cm}^{-3}$ and $\rho_H = 258.6 \text{ cm}^{-3}$, for the free electrons and the hydrogen atom fields, respectively; the radiation field of the cosmic microwave background (CMB) with density $\rho_{\text{CMB}} = 5.319 \times 10^{11} \text{ cm}^{-3}$ and energy $E_{\text{CMB}} = 0.004 \text{ eV}$; and a homogeneous magnetic field $B = 10^{-16} \text{ G}$.

2.1. Characteristic Timescales

In Fig.1 we show the comparison between the cooling times (dashed lines), and the ionization equivalent times (i.e. the time that it takes for a cooling process to drain 13.6 eV from an electron; dotted lines), for each interaction. We considered solely the processes that operate within the energy range of galactic CR ($E \lesssim 1 \text{ PeV}$, i.e. below the knee of the observed energy spectrum). These

being: collisional ionization (CI) (black), Coulomb interaction (red), inverse Compton (IC) scattering (yellow), Bremsstrahlung (green) and synchrotron radiation (blue), for electrons.

We notice that the IC scattering with CMB photons and the CI seem to be the only efficient cooling mechanisms. For other interactions, it would take a time greater than the Hubble time to drain a significant fraction of the electron kinetic energy K_e . Nonetheless, high energy electrons only need to deposit a small portion of their kinetic energy into its surroundings to make a relevant change in the thermodynamic state of the IGM. For this reason, we found necessary to consider other characteristic timescales in order to have a complete understanding of the energy deposition phenomena.

Looking at the ionization equivalent times, we see that although the Coulomb interaction is not an efficient cooling mechanism for electrons, the energy depleted by it can be up to several times the hydrogen ionization potential $I_H = 13.6 \text{ eV}$. In fact, even though the energy lost through this interaction is a small fraction of the total K_e , for $K_e > 10 \text{ keV}$ these losses would be equivalent to thousands of ionization events per primary electron.

In Fig.2 we show the mean interaction times for photons. In this case, that indicator is sufficient since these interactions imply the annihilation of the incident photon. We considered the following processes: photo-ionization (black dash-dotted line), Compton scattering (yellow dash-dotted line) and pair creation (green dash-dotted line). Only photo-ionization for photon energies up to $\sim 1 \text{ keV}$ is relevant since at higher energies, and for other processes, the timescales are greater than the Hubble time.

From the precedent analysis we conclude that the relevant interactions for the purposes of this work are the following: CI, Coulomb interaction, IC scattering, and photo-ionization produced by secondary photons.

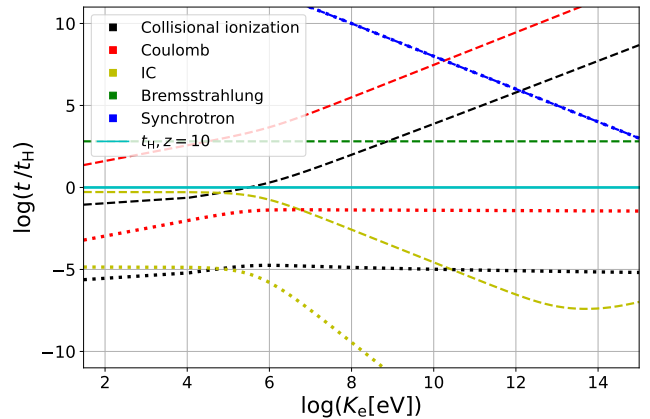


Fig. 1. Energy-loss timescales normalized to the Hubble time for CR electrons. Cooling times (dashed lines) and ionization equivalent times (dotted lines) as functions of the kinetic energy of the electrons.

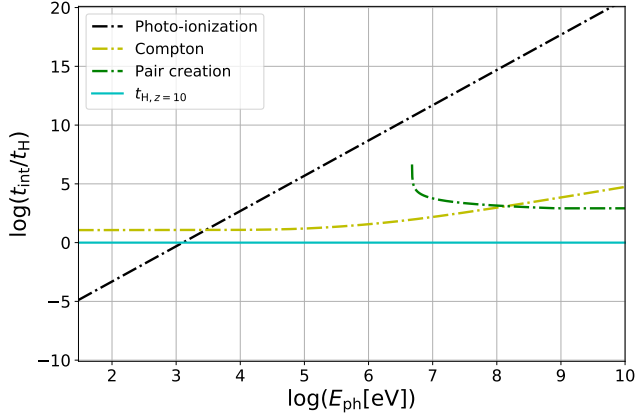


Fig. 2. Photons interaction times normalized to the Hubble time as a functions of their energy (dash-dotted lines).

2.2. Numerical simulations

The evolution and propagation of an electromagnetic cascade involves the coupling of different types of particle populations. This renders the problem too complex to be properly solved by analytic means. Therefore, to model the interactions between CRs and the IGM we used the code UTOPIA (Understanding Transport Of Particles In Astrophysics; Pellizza et al., 2010; Douna et al., 2018). We ran a set of physical simulations spanning a spherical volume of radius $R = 1$ Mpc (typical distance between galaxies at $z = 10$) centered around a point-like source of CR electrons, considering only the cooling processes we found to be relevant for reionization.

3. Preliminary Results

As shown in Fig.3, we calculated the cumulative number of ionization events per primary electron, produced by electron-initiated EMC, as a function of the distance to the source and the kinetic energy of the primary electron. For primary electrons with $K_e < 10$ keV we see that the curves reach a plateau once all the electrons in the cascade cool down below I_H . On the other hand, for electrons with $K_e > 10$ keV, the ionization counts keep increasing for all distance bins, indicating that they do not cool completely before reaching the spatial boundary of the simulation (1 Mpc). These results agree with those of Douna et al. (2018) for low energy electrons, and extend them to higher kinetic energies. They suggest that high energy electrons can transport a fraction of their energy far away from the CR source, opening a potential channel of deep IGM ionization, as we show in the remainder of this section.

In Fig.4 we show a simulated spectra of secondary photons produced by IC scattering, for different kinetic energy values of the primary electron. We see that secondary ionizing photons are originated when electrons with energies between 15 – 500 MeV scatter against a CMB photon ($E_{\text{CMB}} \sim 4.12 \times 10^{-3}$ eV). Electrons below that energy range produce photons with energies $E_{\text{ph}} < 13.6$ eV (i.e. non-ionizing photons). Mean-

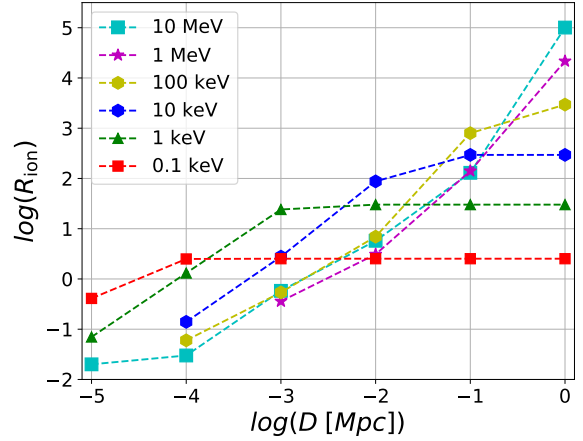


Fig. 3. Ionization count per primary electron (R_{ion}) as a function of the distance D from the source, for different kinetic energy values of the primary electrons.

while, electrons with higher energy produce photons with $E_{\text{ph}} > 1$ keV. These have a mean free path larger than the size of the simulated volume and we therefore disregard them. In conclusion, EMC initiated by electrons with $15 < K_e < 500$ MeV are the ones expected to be of special relevance, due to the contribution of photo-ionization.

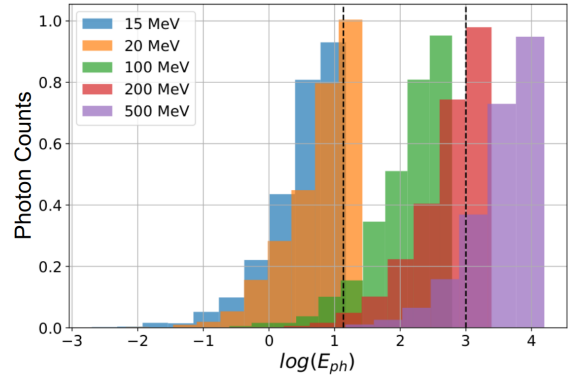


Fig. 4. Normalized energy spectrum of the secondary photons produced by primary electrons via IC. The colors show different kinetic energies for the primary electrons. The two dashed vertical lines limit the region of interest for photon energies, from 13.6 eV (left) to 1 keV (right).

In Fig.5 we show the total number of ionization events per primary electron N_{ion} produced within D , as a function of the electron kinetic energy, for different sets of simulations. The first one includes only elastic and inelastic collisions (dashed blue line); the second one considers also IC interactions (dashed black line); and the third one additionally incorporates the photo-ionization of secondary photons (dashed red line). The plot distinguishes three energy ranges: low (green), intermediate (yellow) and high (red) kinetic energy regions.

In the leftmost region there is no significant difference among the curves, N_{ion} grows linearly with the en-

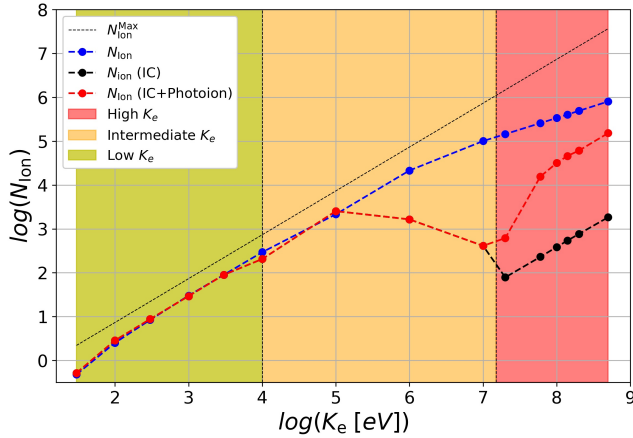


Fig. 5. Ionization counts per primary electron with and without considering IC and photo-ionization contributions, as a function of the kinetic energy of the primary electrons.

ergy of the electron for all cases. First, low energy electrons lose most of their energy via CI, the rate of which heavily dominates over IC scattering at this regime. Second, the mean energy lost in each IC interaction is well below I_H , and only generates non-ionizing secondary photons. Therefore, the effects of IC and photo-ionization on N_{ion} for low energy electrons is negligible.

The middle region shows how the IC losses start to hinder the number of ionization events per primary electron. This occurs because the amount of energy lost with each interaction increases with the energy of the incident electron. For electrons with $K_e > 10$ keV, the energy fraction lost through IC dominates over other losses. This tendency continues as we consider higher electron energy values, until we reach $K_e = 15$ MeV (high energy regime). Then, for intermediate energy electrons, the incorporation of the IC scattering results in a significant decrease of the number of CI events they can produce.

Finally, electrons with $K_e > 15$ MeV can produce, through IC scattering, secondary photons with energies above the ionization threshold. This allows photo-ionization to occur, permitting a significant portion of the energy losses due to IC to be reinserted into the medium as ionizations. This means that high energy electrons can produce a significant contribution to the ionization of the medium, specially on those regions far from the source where low energy ones cannot reach.

4. Conclusions

We explored the physics of electron-initiated EMC propagating through the IGM for a standard reionization scenario that is consistent with the latest observational results (Wise, 2019). To this aim, we simulated all energy deposition mechanisms for electron cooling within a spherical volume of 1 Mpc centered on the CR source, and determined their effects over the IGM during the EoR. Summarizing:

- (i) Both low energy ($K_e < 100$ keV) and high energy ($K_e > 15$ MeV) electrons can ionize efficiently.
- (ii) Intermediate energy electrons ($15 \text{ MeV} > K_e > 100$ keV) cool down rapidly via IC scattering, producing non-ionizing photons, which hinders their ionization efficiency.
- (iii) Escaping electrons carry away a large fraction of their initial energy, and are of interest as ionizing agents of the deep IGM.
- (iv) As suggested by previous works (Douna et al., 2018), the large scale reionization of the IGM mainly proceed in two steps: i) energy has to be carried far away from the source by high energy electrons and ii) those electrons have to be cooled to produce ionizations.

Our future endeavours will focus on extending the spatial range of the simulations in order to have a better understanding of the reionization process beyond the megaparsec threshold. We are also working on the effects of magnetic fields over the propagation of the EMC, since the irregularities in these fields would give rise to a diffusive transport regime for the electrons, which could significantly impact their ionization capability. Additionally, we aim to incorporate hadronic-initiated cascades into our simulations to study the contribution of CR protons to the reionization process. This will allow us to study more physically accurate injections of CR particles, taking into account all relevant energy losses, and then scale the results to the demographics of the CR sources, as was done Douna et al. (2018).

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